Week 3 — Critical Thinking & Data Investigation

Brian - #1, #2, and #3
Isaiah - #4
Rachel - #5

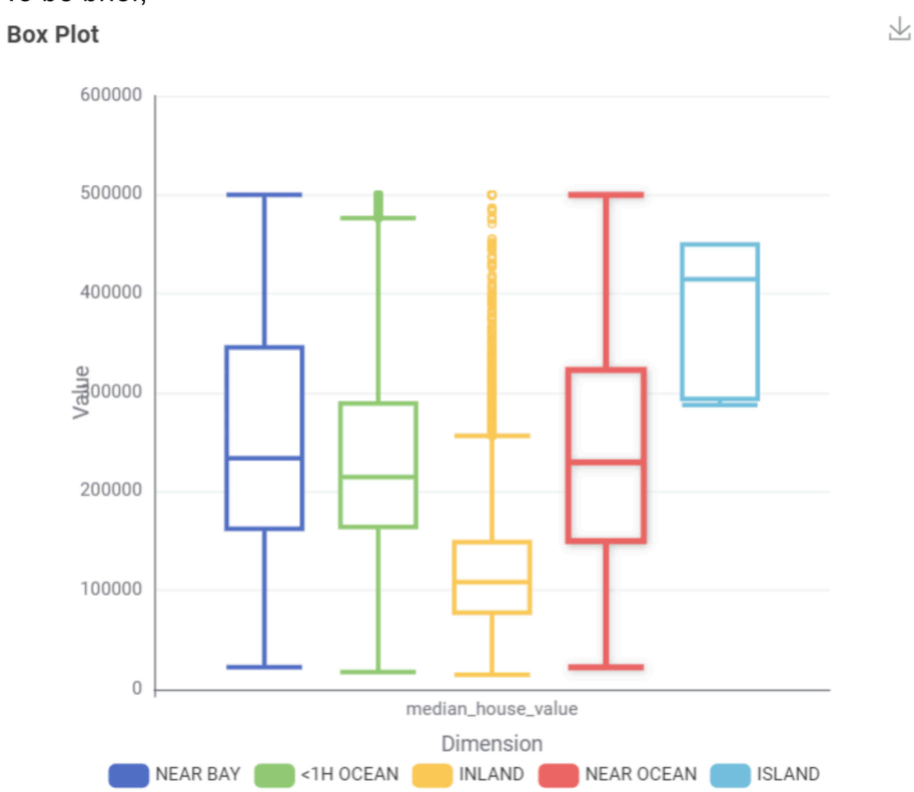
Deliverable 3: 5-Question Analytical Report. Solve the following mysteries using labelled Metanodes and Annotations:

1. **Ocean Proximity:** Is it a unique driver or a proxy for income? (Box Plots/Scatter Plots). Nodes: GroupBy, Box Plot, Scatter Plot, Annotation. Form your hypothesis in an Annotation before running anything, argue your conclusion in a second Annotation after.

This is my current hypothesis: This question really depends. Ocean proximity appears to influence housing prices but it may not always be a direct driver. It could act like a proxy since waterfront/coastal areas typically have higher median incomes.

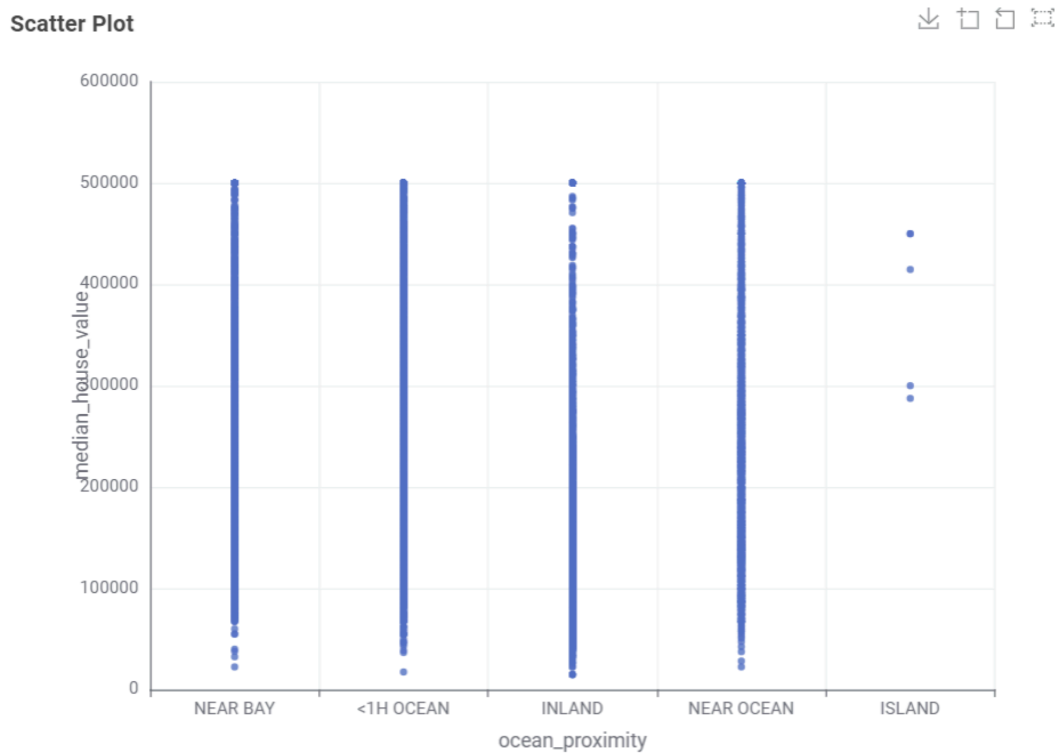
When we look at the data, I am going to use a box plot, scatterplot, and histogram. They are all looking at the median housing value and the ocean proximity.

To be brief,



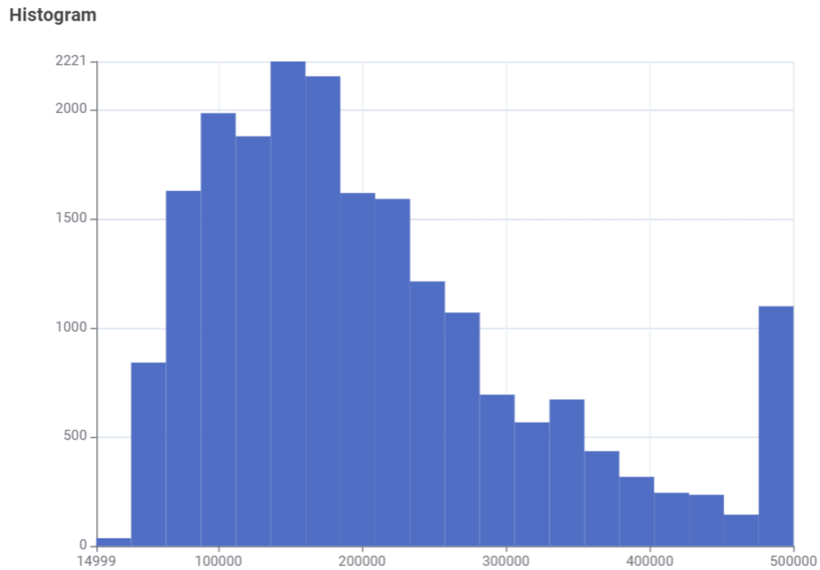
1. Box Plot
    a. Inland: shows the lowest median housing values at $108500
    b. Island: shows the highest median values (but smaller dataset)
    c. Near ocean: has the highest median (besides island) at $229450
    d. Near bay: has the 3rd highest overall median at $233800
    e. <1H Ocean: 4th highest median at $214850

2. Scatter plot



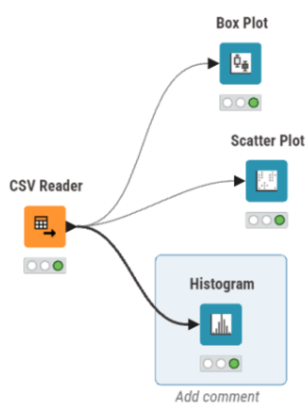
This graph provides little insight. What we can say is that housing prices may have a trend but overall are not totally reliant on the area. For example, even though our box plot and scatter plot show that inland properties have a lot of cheaper housing, they still have their fair share of really expensive homes (>$500,000). It looks like there are not many island housing property options, but all of them are typically more expensive than all other areas. This graph concludes that all areas have poor areas, and all areas have rich areas (besides island properties). A better showcase of this would be the boxplot that is above.

3. Histogram



The histogram was just to give a general idea of all the housing provided in the dataset. Our analysis here shows that the large majority of housing (regardless of area), are in between $100,000 and $300,000. There are outliers with homes less than $100,000 and housing above $400,000. This graph could also be a precursor to showing that specific areas may contain more expensive housing based on the area.

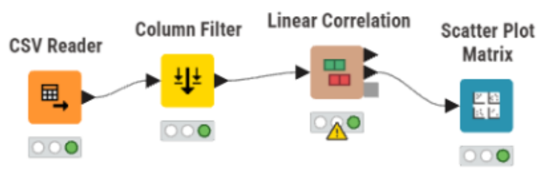
Overall, the California neighborhoods/areas does seem like it can be a proxy for housing prices, but there is a caveat that all areas do have expensive housing and all areas do have cheap housing. It just happens to be that specific areas have higher median housing prices and have more high end options than others. This is highly supported by our box plot and scatter plot graphs.



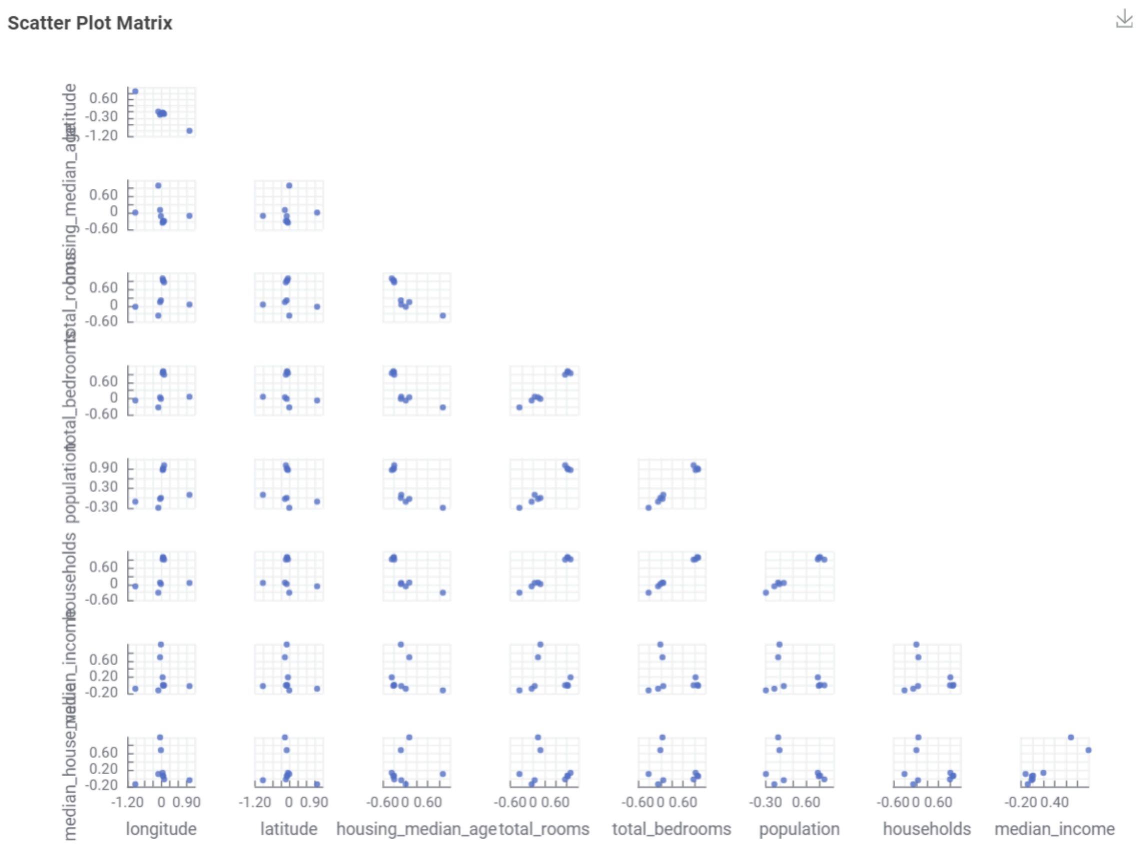
2. **Redundancy:** Which variable would you remove from the model based on **Linear Correlation**?

   Nodes: Linear Correlation, Scatter Matrix, Column Filter, Annotation. Pick one variable, justify it with the correlation values — not intuition.

To check all the correlations, I created a 4 node knime chain. CSV reader -> column filter (filtered out proximity because its not a number), linear correlation -> scatter plot matrix.



This organization allows me to see all the scatter plots between every variable to see how correlated each pair of attributes are in one snapshot.



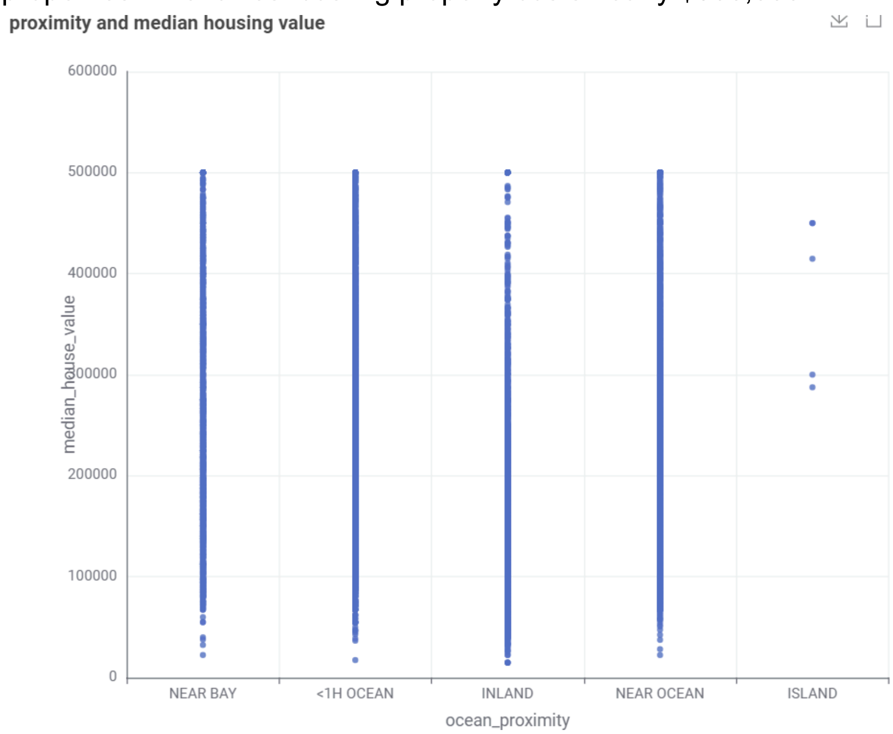
The scatter plot matrix shows strong linear relationships with a lot of variables, particularly the ones on the right of the matrix. The strong linear correlation is shown between total_rooms, total_bedrooms, households, and population. These all show the tightest linear patterns which indicates strong correlation and redundancy.

The obvious variable to remove is the total_bedrooms. This is because of its constant redundancy. In the scatter plot matrix, there is a very strong linear relationship between the total_bedroms and households which tells us that they contain identical information. This also just makes logical sense … Most people don't buy a 3 bedroom house for a family of 5 people.

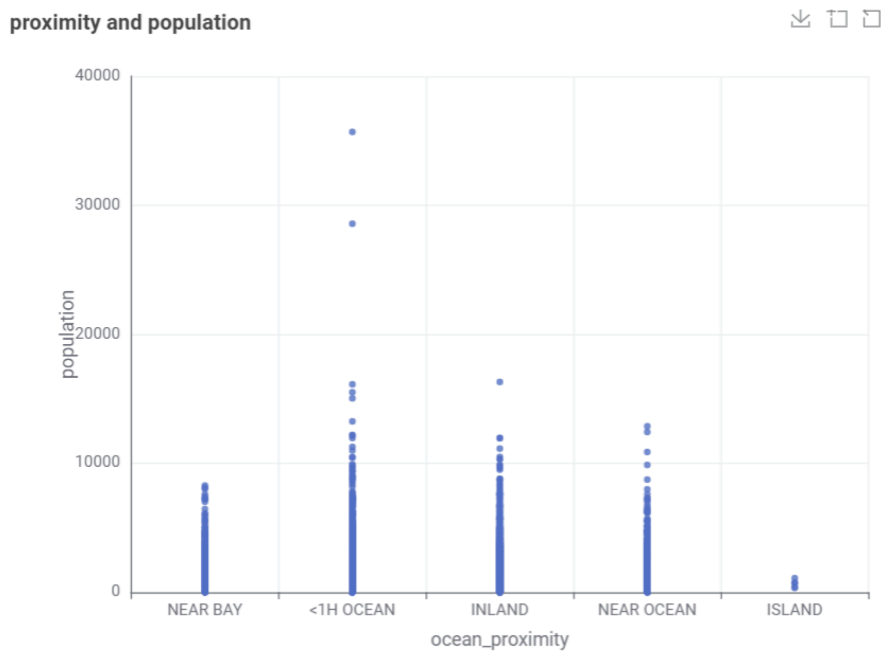
The correlation is strong it looks like it could be very close to value of 1. This means that total_bedrooms is not providing a unique or unexpected predicted value. Removing it will allow us to lose minimal information while also simplifying the model .

3. **Outliers:** Pinpoint a neighborhood that is a geographic outlier and speculate on real-world causes. Nodes: Scatter Plot, Row Filter, Statistics, Annotation. Speculate on real-world causes in the Annotation.

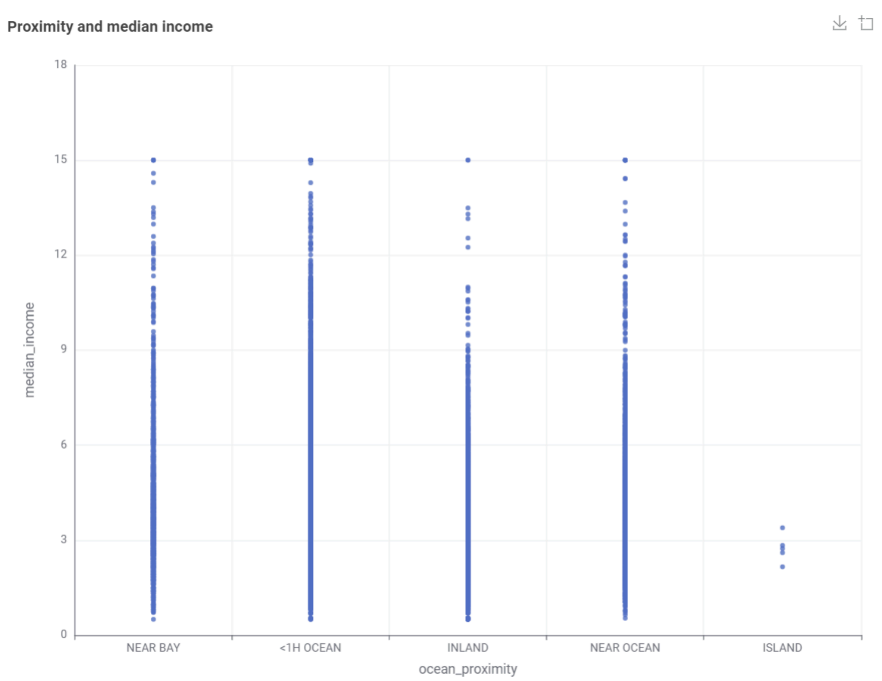
To conduct this analysis, I used 3 scatter plots to help make reasonable assessments about real world causes on a geographic outlier. Prior work (question 1) shows that the geographic outlier in terms of housing is the island properties. The island area was promptly decided as an outlier in terms of price due to it having no low cost (<$100,000) properties and only higher cost properties. The lowest costing property costs nearly $300,000.



First, we need to confirm our hypothesis that the island properties are a geographic outlier. The heatmap above shows all the properties and the average housing value. Clearly, we can see that the assessment made above is supported by the graph and shows that island properties have very few values and they also have very expensive values on average. A speculation that we can make is that there are very few properties available for island housing. This can be further noted by the idea of supply and demand. Island properties are known to be very beautiful and hard to get housing materials to making them have a small supply and a lot of wishful demand.

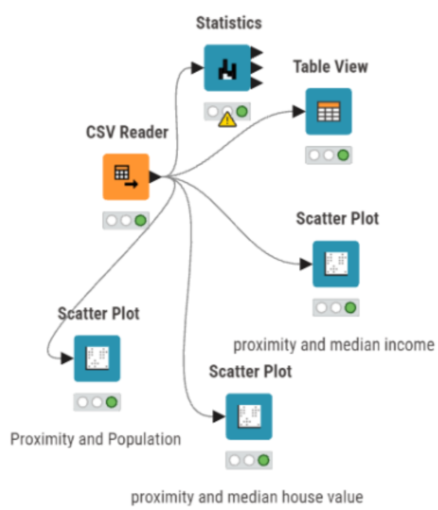


The graph above is used to assess the amount of people living in each neighborhood area. Previously, I mentioned the concept of supply and demand. The population and proximity graph was used to estimate how many people the island neighborhoods can sustain. This graph shows a good estimate of the population of each area. If we look at the neighborhoods with less people, we can see that those areas typically have higher median costs of living. A reasonable assumption on why the island neighborhood is such an outlier is simply because there may not be enough geographical space to support the amount of people and housing that may be ideal.



Lastly, we have a scatter plot showing proximity and median income. The graph here is interesting because while the island neighborhoods have the highest average median housing cost, they have by far the lowest average median income. This is a phenomenon that can be noted in many areas of the United States. While this doesn't tell us more information about why the islands are a financial cost outlier, it can point to a pattern we commonly see in the United States between expensive housing and land. This is common in North Carolina, retired citizens sometimes own expensive homes, lots of land, and have little to no income in desirable areas. This perspective may be applicable to those living on the island, retired and not working. This can also be supported by the fact that island neighborhoods may not have a lot of infrastructure which can equate to little economic drive and job opportunities.

In conclusion, the neighborhood that is the outlier is the Island neighborhood. This is supported by statistical evidence from question 1, and also in the scatter plots above. The reason these may be the most expensive housing is the ideal location and the idea of the supply and demand that these islands face. Islands are typically favorable locations, but have little geographic space to house people.



4. **Regional Logic:** Is the income-to-value relationship identical in Northern and Southern California?

   Nodes: Row Splitter, Linear Correlation, Scatter Plot, Line Plot, Annotation. Split on latitude > 37.5 for North/South.

Overall the correlation between both income and value is strong and positive for both Northern and Southern California. Overall, Southern California has a lightly higher correlation at 0.693. Using the household income as a predictor for the value of the home is successful for both regions. Using the Row splitter to run workflows on rows based on certain attributes is a great way to determine information about homes.

Income-to-value for Northern California:

Rows: 1 | Columns: 5

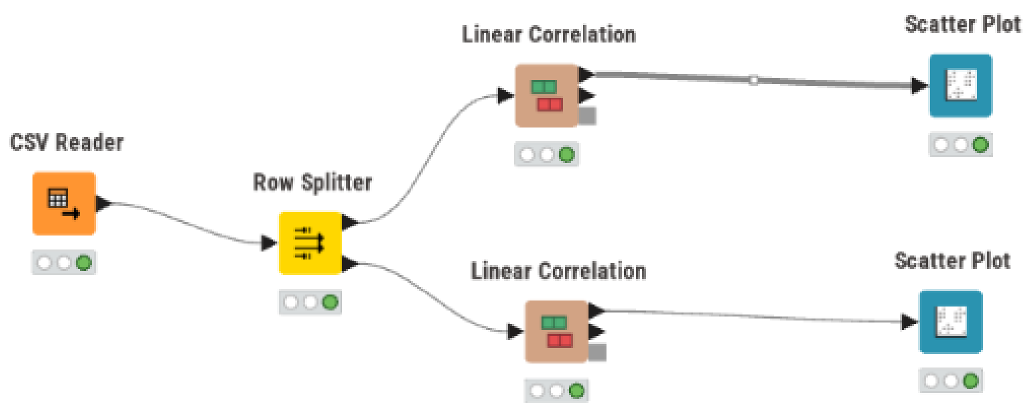
#	RowID	First column name (String)	Second column name (String)	Correlation value (Number (Float))
1	Row0	median_income	median_house_value	0.668

Income-to-value for Southern California:

Rows: 1 | Columns: 5

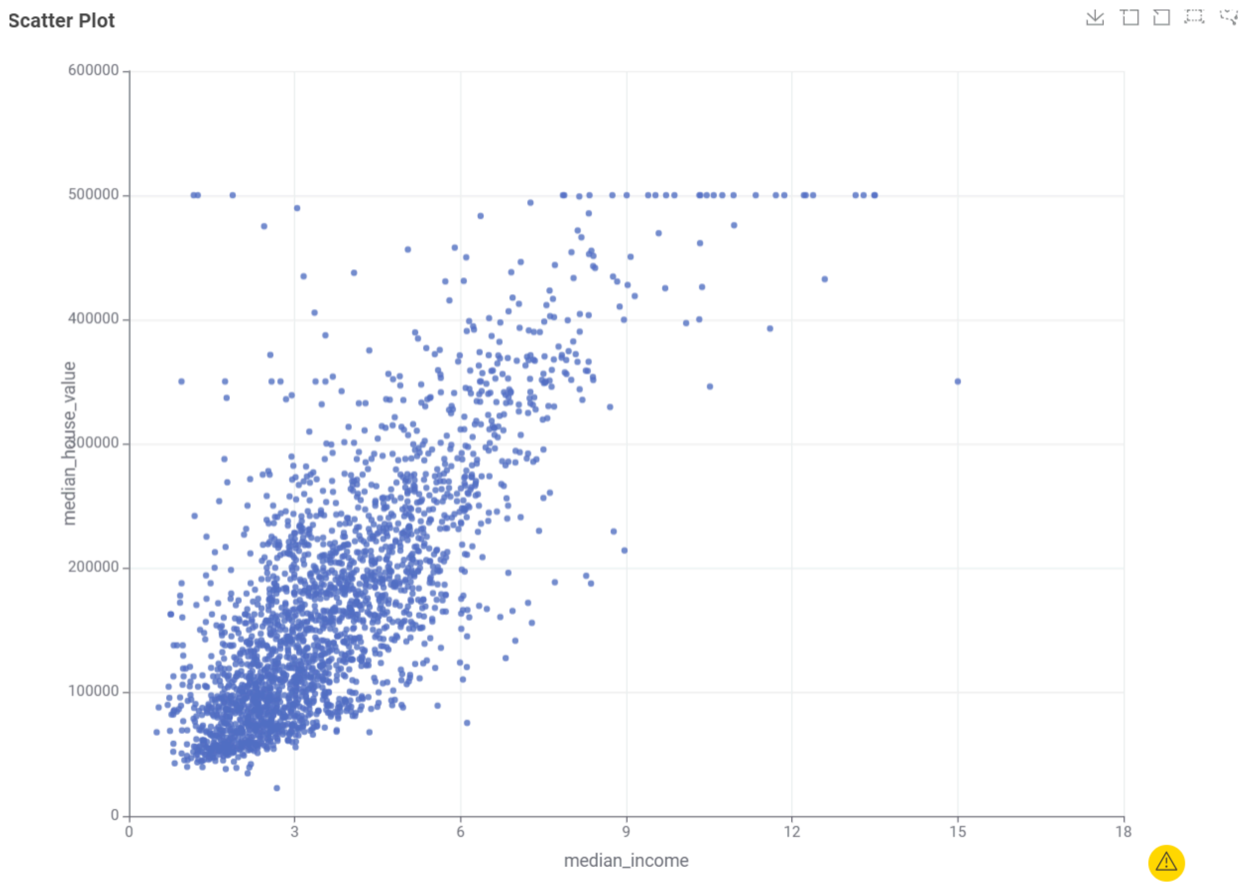
#	RowID	First column name (String)	Second column name (String)	Correlation value (Number (Float))	p value (Number (Float))
1	Row0	median_income	median_house_value	0.693	0

Workflow:



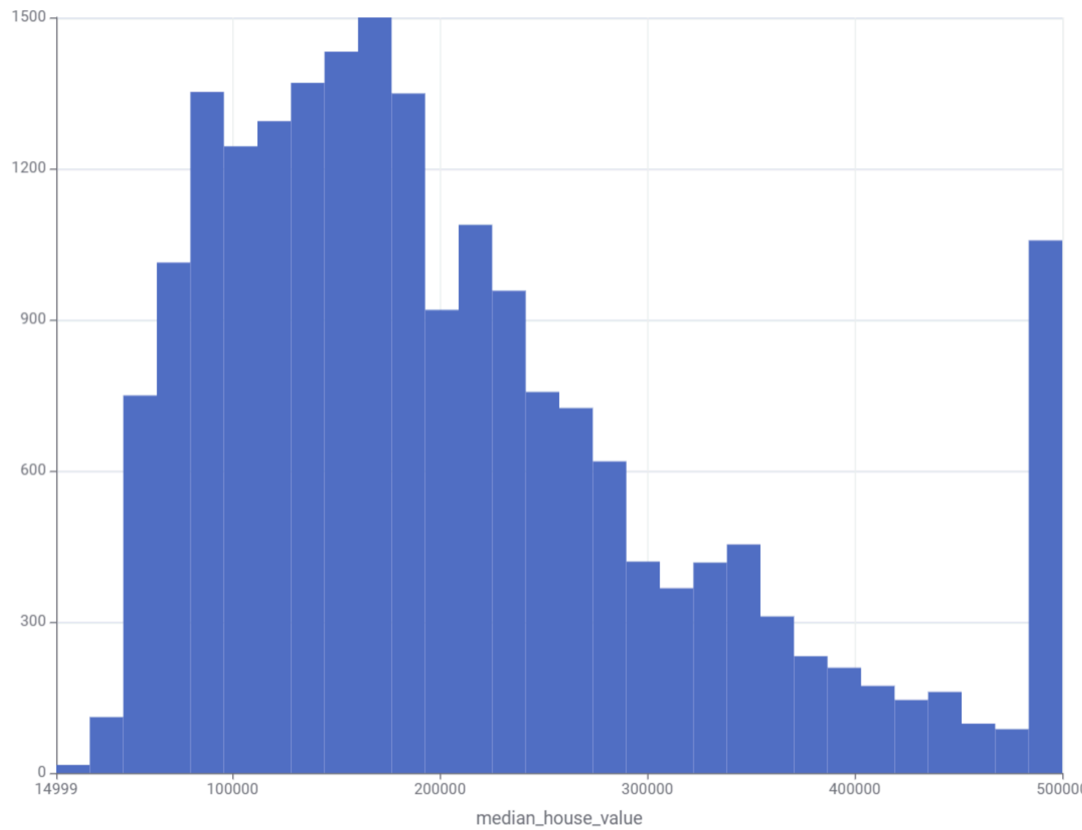
5. **Market Caps:** At what income level does the house value "flatline," and is this a data artifact?
   Nodes: Scatter Plot, Row Filter, GroupBy, Histogram, Annotation. Explain whether the cap is a real market phenomenon or a data artifact.

To determine where housing values flatline, I created a scatter plot comparing the median income and median house value. The scatter plot shows that housing prices increase with income until approximately a median income of 8-9. After this point, the values stop increasing and cluster around $500,000.



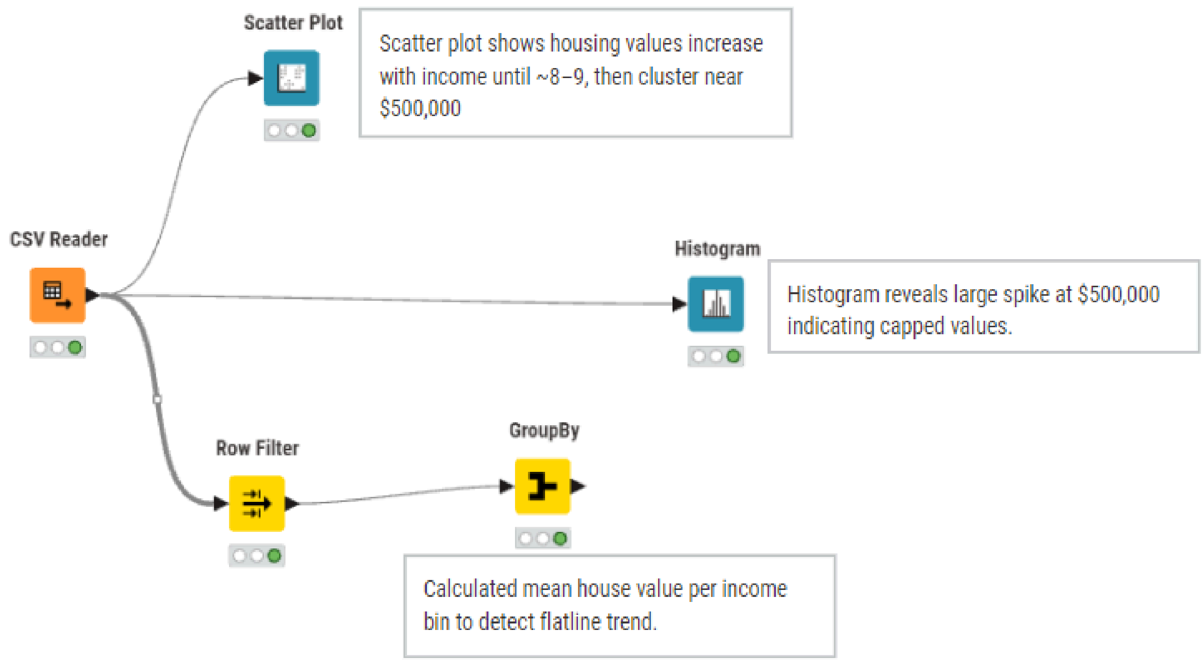
To further investigate, I grouped the data using income bins and calculated the mean housing value. The GroupBy results confirmed that higher income bins consistently produce average values close to $500,000. A histogram of median house value also shows a noticeable spike at exactly $500,000.

Histogram



The pattern suggests that the flatline is not a natural market behavior. Instead, it appears to be a data artifact caused by a cap in the dataset. Many observations share the exact same minimum value, which indicates that housing values above $500,000 were likely truncated.

Overall, the house value flatline occurs at a median income of approximately 8-9, and this cap is a data artifact rather than a real-world market phenomenon as explained above for the following reasons.



Team Weekly Sync: Project Checklist

Group Name/Number: Group 5 | **Date:** 04/04/2026

Use this checklist at the start of every meeting to ensure you are meeting the "Green-to-Gold" requirements and project milestones.

1. Infrastructure & Accountability (Start-of-Meeting)

- [✔] **GitHub Sync:** Did everyone pull the latest changes from `main`?

- [✔] **Contribution Log:** Has the `README.md` been updated to reflect progress made this week?
- [✔] **Blocking Issues:** Is anyone currently "blocked" by a KNIME node error or a data issue? *(If yes, resolve this first!)*

2. Current Sprint Focus (Check the Week)

- **Week 3 (Prep):** Is your feature matrix complete?

3. Technical Quality Assurance (The "Green-to-Gold" Audit)

- [✔] **Automation:** Does the workflow run from `File Reader` to `Output` without changing settings?
- [✔] **Annotation:** Are your primary logic steps (e.g., normalisation, optimization) explained in a **Metanode Annotation**?
- [✔] **Evidence:** Is every chart/table in your report directly linked to a specific KNIME node output?

4. Action Items (End-of-Meeting)

Who is doing what before next week?

Task	**Assignee**	**Deadline**
Reviewing next assignment	Brian	4/8
Reviewing next assignment	Isaiah	4/8
Reviewing next assignment	Rachel	4/8